

Preface

At the root of measure theory is a rigorous treatment of how to extend the notions of length (on the real line), area (on the Euclidean plane), and volume (in three-space) to a larger collection of sets than those seen in, say, courses on Calculus. We have seen in a first course in analysis that there exist sets with some exceptionally strange properties (the Cantor set being a classic example). We would like to be able to discuss these types of sets in the context of measuring their sizes (be it a length, area, volume, etc). While the historical impetus was to solve this problem for sets in $\mathbb{R}^n$, one finds that it is not hard to abstract the theory to more general sets. Accordingly we take up this abstract study. This is not just for the sake of abstraction. In fact one can find applications for abstract measure theory in, for example, probability theory (where the measure of a set corresponds to the likelihood of an element in that set being chosen under some random process) and physics (where one might want the measure of a set to correspond to the mass of some object occupying that space).

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Chapter 1

Preliminaries

§ 1.1. Set Theory

Definition 1.1.1. A *set* is a collection of different things; the things are called *elements* of the set.

Zermelo–Fraenkel set theory introduces eight different axioms which mathematics builds upon. Below is an introduction to the ones which are relevant for the material covered in this course.

Axiom 1 (Axiom of Extension). *If the sets X and Y have the same members, then they are the same set. In other words:*

$$(\forall X)(\forall Y)\left((\forall z)(z \in X \Leftrightarrow z \in Y) \implies X = Y\right)$$

Axiom 2 (Axiom of Existence). *There is a set such that no element is a member of it. In other words:*

$$(\exists A) : (\forall x)(x \notin A)$$

Note that the Axiom of Extension immediately proves that the set which contains no elements is unique; two sets are equal if they have the same members, hence two sets which contain no elements are equal.

Definition 1.1.2. The set which contains no elements is called the *empty set*, and is denoted by $\emptyset$.

Axiom 3 (Axiom of Pairing). *Let A and B be sets. There exists the set $\mathcal{C} := \{A, B\}$. In other words:*

$$(\forall A)(\forall B)(\exists \mathcal{C}) : (A \in \mathcal{C} \wedge B \in \mathcal{C}).$$

Axiom 4 (Axiom of Union). *Let $\mathcal{C}$ be a set. Then there exists a set U consisting of the elements x that are contained in at least one element of the set $\mathcal{C}$, that is:*

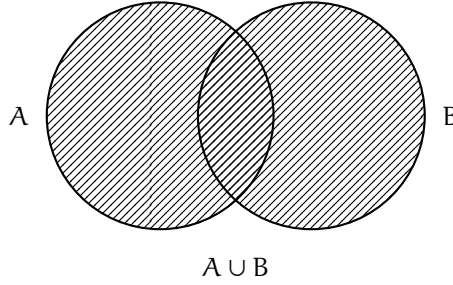
$$(\forall \mathcal{C})(\exists U) : (\forall x)(x \in U \Leftrightarrow (\exists C)(u \in C \wedge C \in \mathcal{C}))$$

Definition 1.1.3. Let $\mathcal{C}$ be a set, and let U be the set consisting of the elements x that are contained in at least one element of $\mathcal{C}$.

(1) The set U is called the *union of the set $\mathcal{C}$* and is denoted:

$$U := \bigcup \mathcal{C}.$$

(2) If the set $\mathcal{C}$ consists of only two sets A and B , then we write $U := A \cup B$.



Axiom 5 (Axiom Schema of Specification). *Let A be a set, and let φ be a predicate containing the free variable x . There exists a subset B of the set A whose members are precisely the elements x of the set A such that the sentence $\varphi(x)$ is true. In other words:*

$$(\forall A)(\forall \varphi)(\exists B) : (B = \{x \in A \mid \varphi(x)\})$$

Proposition 1.1.1. *Let $\mathcal{C}$ be a nonempty set. There exists a set D consisting of the elements x that are contained in all elements of the set $\mathcal{C}$.*

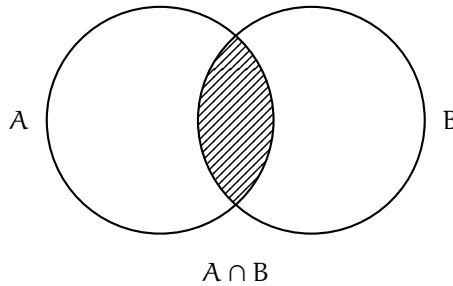
Proof. Let $E \in \mathcal{C}$. By the Axiom Schema of Specification we have $D := \{c \in E \mid (\forall X)(X \in \mathcal{C} \Rightarrow c \in X)\}$. $\square$

Definition 1.1.4. Let $\mathcal{C}$ be a nonempty set, and let D be the set consisting of the elements x that are contained in all elements of the set $\mathcal{C}$.

(1) The set D is called the *intersection of the set $\mathcal{C}$* and is denoted:

$$D := \bigcap \mathcal{C}.$$

(2) If the set $\mathcal{C}$ contains only two sets A and B , then we write $D := A \cap B$.



Axiom 6. *Let X be a set. Then the set of all subsets of X exists.*

Definition 1.1.5. Let X be a set. Then the set of all subsets of the set X is called the *power set* of the set X and is denoted $\mathcal{P}(X)$.

Axiom 7 (Axiom of Infinity). *There is a set that contains all natural numbers.*

Definition 1.1.6. An *index set* is a nonempty set whose elements label the elements of another set. We denote the collection of objects labeled by elements of I as $\{A_\alpha\}_{\alpha \in I}$.

With this definition and the Axiom of Infinity, we can extend the concept of unions and intersections to families of sets of arbitrary size.***

Definition 1.1.7. Let I be an index set, let $\{X_\alpha\}_{\alpha \in I}$ be an indexed family of sets, and write $\mathcal{X} := \{X_\alpha \mid \alpha \in I\}$. Define:

$$\bigcup_{\alpha \in I} X_\alpha := \bigcup \mathcal{X},$$

$$\bigcap_{\alpha \in I} X_\alpha := \bigcap \mathcal{X}.$$

Definition 1.1.8. *** Let $\{X_n\}_{n \in \mathbb{N}}$ be a family of sets indexed by the natural numbers.

(1) The *limit superior* of $\{X_n\}_{n \in \mathbb{N}}$ is:

$$\begin{aligned} \limsup X_n &:= \bigcap_{k=1}^{\infty} \left(\bigcup_{n=k}^{\infty} X_n \right) \\ &= \{x \mid x \in E_n \text{ for infinitely many } n\}. \end{aligned}$$

(2) The *limit inferior* of $\{X_n\}_{n \in \mathbb{N}}$ is:

$$\begin{aligned} \limsup X_n &:= \bigcup_{k=1}^{\infty} \left(\bigcap_{n=k}^{\infty} X_n \right) \\ &= \{x \mid x \notin E_n \text{ for only finitely many } n\} \end{aligned}$$

Definition 1.1.9. Let A and B be two sets. Define the *difference* of A and B by $A \setminus B := \{x \in A \mid x \notin B\}$.

Note that the difference of two sets exists by the Axiom Schema of Specification.

Definition 1.1.10. Let a and b be two objects. The *ordered pair* (a, b) is defined by $(a, b) := \{\{a\}, \{a, b\}\}$.

Proposition 1.1.2. Let A and B be sets. There exists a set E containing all ordered pairs (a, b) , where $a \in A$ and $b \in B$.

Proof. Let (a, b) be any ordered pair such that $a \in A$ and $b \in B$. By definition $(a, b) = \{\{a\}, \{a, b\}\}$. Now $\{a\} \subset A$, so $\{a\} \subset A \cup B$, hence $\{a\} \in \mathcal{P}(A \cup B)$. Similarly, $\{a, b\} \subset A \cup B$, so $\{a, b\} \in \mathcal{P}(A \cup B)$. It follows that $\{\{a\}, \{a, b\}\} \subset \mathcal{P}(A \cup B)$, hence $\{\{a\}, \{a, b\}\} \in$

$\mathcal{P}(\mathcal{P}(A \cup B))$. We know all these sets are known to exist by the Axiom of Union and Axiom of Power Set. Then by the Axiom Schema of Specification, there exists a set:

$$E := \{t \in \mathcal{P}(\mathcal{P}(A \cup B)) \mid (\exists a)(\exists b)(a \in A \wedge b \in B \wedge t = (a, b))\}.$$

Moreover, this set is unique by the Axiom of Extension, and is the set of all ordered pairs (a, b) with $a \in A$ and $b \in B$. $\square$

Definition 1.1.11. Let A and B be two sets, and let E be the set containing all ordered pairs (a, b) where $a \in A$ and $b \in B$. The set E is called the *Cartesian product of A and B* , and is denoted $E := A \times B = \{(a, b) \mid a \in A \wedge b \in B\}$.

Definition 1.1.12. Let X and Y be sets. A *relation* from X to Y is a subset R of $X \times Y$. We write xRy to mean $(x, y) \in R$.

- (1) An *equivalence relation* on X is a relation which is reflexive, transitive, and symmetric. We write $x \sim y$ to mean xRy .
- (2) A *function* (or *map*) from X to Y , denoted $f : X \rightarrow Y$, is a relation from X to Y such that, for every $x \in X$, there exists a unique $y \in Y$ such that xRy . We write $f(x) = y$ to mean xRy .
- (3) A *partial order* on X is a relation which is reflexive, transitive, and antisymmetric. We write $x \leq y$ to mean xRy , and say that X is an *ordered set*.

Proposition 1.1.3. *** Let X be a set. The following are equivalent:

- (1) S is a partition of X ;
- (2) There exists an equivalence relation $\sim$ on X such that the equivalence classes of $\sim$ are exactly the elements of S .

Proof. $\square$

Example 1.1.1. An equivalent relation on X partitions X into equivalence classes.

Example 1.1.2. Recall that, if $\mathcal{C}$ and $\mathcal{D}$ are categories, then a *covariant functor* $T : \mathcal{C} \rightarrow \mathcal{D}$ is a map such that:

- (i) If $A \in \text{obj}(\mathcal{C})$, then $T(A) \in \text{obj}(\mathcal{D})$.
- (ii) If $f \in \text{Hom}_{\mathcal{C}}(A, A')$, then $T(f) \in \text{Hom}_{\mathcal{D}}(T(A), T(A'))$.
- (iii) If $f \in \text{Hom}_{\mathcal{C}}(A, A')$ and $g \in \text{Hom}_{\mathcal{C}}(A', A'')$, then $T(f) \in \text{Hom}_{\mathcal{D}}(T(A), T(A'))$ and $T(g) \in \text{Hom}_{\mathcal{D}}(T(A'), T(A''))$. In particular:

$$T(gf) = T(g)T(f).$$

- (iv) For every $A \in \text{obj}(\mathcal{C})$, $T(\text{id}_A) = \text{id}_{T(A)}$.

A *contravariant functor* $T : \mathcal{C} \rightarrow \mathcal{D}$ is similarly defined, but (ii) and (iii) are instead defined as:

- (ii) If $f \in \text{Hom}_{\mathcal{C}}(A, A')$, then $T(f) \in \text{Hom}_{\mathcal{D}}(T(A'), T(A))$.

- (iii) If $f \in \text{Hom}_{\mathcal{C}}(A, A')$ and $g \in \text{Hom}_{\mathcal{C}}(A', A'')$, then $T(f) \in \text{Hom}_{\mathcal{D}}(T(A'), T(A))$ and $T(g) \in \text{Hom}_{\mathcal{D}}(T(A''), T(A'))$. In particular:

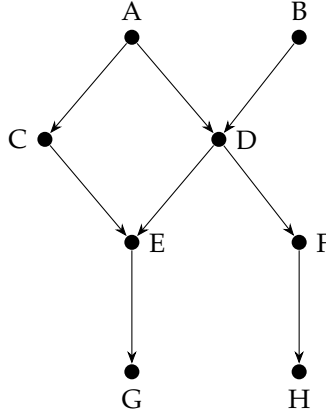
$$T(gf) = T(f)T(g).$$

Given $f : X \rightarrow Y$, the power set gives rise to two functors, the *contravariant power set functor* $\mathbf{Set}^{\text{op}} \rightarrow \mathbf{Set}$ and the *covariant power set functor* $\mathbf{Set} \rightarrow \mathbf{Set}$. The first sends $f : X \rightarrow Y$ to the *preimage* function $f^{-1} : \mathcal{P}(Y) \rightarrow \mathcal{P}(X)$, whereas the second sends f to the *image* function $f_* : \mathcal{P}(X) \rightarrow \mathcal{P}(Y)$. The preimage is more well-behaved, commuting with boolean operations:

$$\begin{aligned} f^{-1}(E^c) &= (f^{-1}(E))^c, \\ f^{-1}\left(\bigcup_{\alpha \in I} E_{\alpha}\right) &= \bigcup_{\alpha \in I} f^{-1}(E_{\alpha}), \\ f^{-1}\left(\bigcap_{\alpha \in I} E_{\alpha}\right) &= \bigcap_{\alpha \in I} f^{-1}(E_{\alpha}). \end{aligned}$$

This is not necessarily the case for the image function.

Example 1.1.3. *****(I don't like the prose here) Consider the directed acyclic graph:



Let $G = \{A, B, C, D, E, F, G, H\}$ be the set of vertices of our graph. We can define a partial ordering on G as follows: for any $x, y \in G$, we have $x \leq y$ if and only if there exists a directed path from x to y . However, note that not every element of G can be compared.

Definition 1.1.13. An ordering on a set X is said to be *linear* (or *total*) if for every $x, y \in X$, $x \leq y$ or $y \leq x$.

Definition 1.1.14. Let X be an ordered set. Let $A \subset X$.

- (1) A is called *bounded above* if there exists an element $u \in X$ with $a \leq u$ for all $a \in A$. Such a u is called an *upper bound* for A . The set of upper bounds of A is denoted $\mathcal{U}_A = \{u \in X \mid u \text{ is an upper bound of } A\}$.

- (2) A is called *bounded below* if there exists an element $v \in X$ with $v \leq a$ for all $a \in A$. Such a v is called a *lower bound* for A . The set of lower bounds of A is defined as $\mathcal{L}_A = \{v \in X \mid v \text{ is a lower bound of } A\}$.
- (3) If A admits an upper bound u with $u \in A$, then u is called *the greatest element of* A .
- (4) If A admits a lower bound v with $v \in A$, then v is called *the least element of* A .
- (5) If l is the least element of $\mathcal{U}_A$, we write $l = \sup(A)$ and call it the *supremum of* A .
- (6) If g is the greatest element of $\mathcal{L}_A$, we write $g = \inf(A)$ and call it the *infimum of* A .
- (7) A *maximal element of* A is an element $m \in A$ such that if $a \geq m$, then $a = m$ (not necessarily unique).
- (8) A *minimal element of* A is an element $n \in A$ such that if $a \leq n$, then $a = n$ (not necessarily unique).

Definition 1.1.15. If a linear order on a set X satisfies the property that every nonempty subset has a minimal element, then we say it is a *well-ordering*.

—————/—————

***include stuff HERE

Definition 1.1.16.

do general cartesian product
 question is it nonempty?
 axiom of choice
 answer: no, axiom of choice says this choice function exists.
 'shoes vs. socks' formulation of the axiom

—————/—————

§ 1.2. The Extended Real Numbers***

Definition 1.2.1. The *extended real number line* is the set $\overline{\mathbb{R}} := \mathbb{R} \cup \{\infty, \infty\} = [-\infty, \infty]$.

The extended real number line has many unique properties.

- For all $x \in \mathbb{R}$, we have $-\infty < x < \infty$. This allows us to extend the ordering of $\mathbb{R}$ to $\overline{\mathbb{R}}$.
- Recall that since $\mathbb{R}$ is complete, every *bounded subset* admits a supremum. However, for the extended real number line, *every subset* of $\overline{\mathbb{R}}$ admits a supremum.

Example 1.2.1.

- (i) In $\overline{\mathbb{R}}$, we have $\sup \mathbb{R} = \infty$.
- (ii) $\inf \mathbb{Q} \cap (-\infty, 0) = -\infty$.

(iii) $\inf \emptyset = \infty$. This follows from basic intuition: the infimum of a set is the greatest lower bound, so the infimum of the empty set is an element which is a lower bound for nothing. Hence it must be ∞ .

- In $\mathbb{R}$, any sequence which is monotone and bounded converges. However, in $\overline{\mathbb{R}}$, *any* monotone sequence converges to a value in $\overline{\mathbb{R}}$.
- For any sequence $(x_n)_n$, we define:

$$\limsup x_n := \inf_{k \geq 1} \left(\sup_{n \geq k} x_n \right),$$

$$\liminf x_n := \sup_{k \geq 1} \left(\inf_{n \geq k} x_n \right).$$

In $\mathbb{R}$, it could be the case that each is properly divergent, however in $\overline{\mathbb{R}}$ the limit superior and limit inferior *always* exist.

- We can define arithmetic with $-\infty$ and ∞ . For $x \in \overline{\mathbb{R}}$:

$$\begin{aligned} x + \infty &= \infty, \\ x - \infty &= -\infty, \\ \infty + \infty &= \infty, \\ -\infty - \infty &= -\infty. \end{aligned}$$

We don't give any meaning to $\infty - \infty$. If $x > 0$, we define:

$$\begin{aligned} x \cdot \infty &= \infty, \\ x \cdot (-\infty) &= -\infty. \end{aligned}$$

If $x < 0$, we define:

$$\begin{aligned} x \cdot \infty &= -\infty, \\ x \cdot (-\infty) &= \infty. \end{aligned}$$

Unless stated otherwise, we define $0 \cdot \pm\infty = 0$.

- Let X be any set and let $f : X \rightarrow [0, \infty]$. We'd like to give some meaning to $\sum_{x \in X} f(x)$. Regardless of the cardinality of X , it is a theorem that

This whole section needs to be redone.

summations are defined like this: $\sum_{x \in X} f(x) = \sup \{ \sum_{x \in Y} f(x) \mid Y \subset X, Y \text{ finite} \}$.

Proposition 1.2.1. **** (could be cleaned up) Every open set in $\mathbb{R}$ is at most a countable union of disjoint open intervals.*

Proof. Let $U \subset \mathbb{R}$ be open. Define a binary relation $\sim$ on U as follows: $x \sim y$ if and only if there exists an open interval I such that $\{x, y\} \subset I \subset U$. One can check that this is indeed an equivalence relation, hence U is partitioned by the equivalence classes of $\sim$.

We are now going to show that each equivalence class is an open interval. Let E be an equivalence class and let $I = (\inf E, \sup E)$. Clearly $E \subset I$. Let $x \in I$. Then $\inf E < x$ and $x < \sup E$. We can find $u, v \in E$ such that $\inf E < u < x < v < \sup E$. Since $u, v \in E$, we can find an interval J such that $\{u, v\} \subset J \subset U$. But this means $x \in J$, hence $x \sim u$ and $x \sim v$. Thus $x \in E$, giving $I \subset E$ meaning that E is an open interval.

It remains to show that there are at most a countable number of equivalence classes. Note that each equivalence class contains a rational number, and no two classes can contain the same rational number (otherwise the equivalence classes wouldn't be disjoint). By the Axiom of Choice, we can find a choice function $f : U/\sim \rightarrow \mathbb{Q}$ which is injective. Therefore there are at most countably many open intervals. $\square$

§ 1.3. Metric Spaces

Definition 1.3.1. A *distance function* (or *metric*) on a nonempty set X is a map:

$$\rho : X \times X \rightarrow [0, \infty)$$

satisfying for all $x, y, z \in X$:

- (1) $\rho(x, x) = 0$;
- (2) $\rho(x, y) = \rho(y, x)$;
- (3) $\rho(x, z) \leq \rho(x, y) + \rho(y, z)$.

The pair (X, ρ) is called a *metric space*.

Distance functions let us talk about sets which centered at a point.

Definition 1.3.2. Let X be a metric space, $x \in X$ and $r > 0$.

- (1) The *open ball of radius r centered at x* is the set $B(r, x) := \{y \in X \mid \rho(x, y) < r\}$.
- (2) The *closed ball of radius r centered at x* is the set $V(r, x) := \{y \in X \mid \rho(x, y) \leq r\}$.
- (3) The *sphere of radius r centered at x* is the set $S(r, x) := \{y \in X \mid \rho(x, y) = r\}$.

Such distance functions induce a topology on our metric space X .

Definition 1.3.3. Let X be a metric space.

- (1) We say $U \subset X$ is *open* if, for all $x \in U$, there exists $r > 0$ such that $B(r, x) \subset U$.
- (2) We say $V \subset X$ is *closed* if V^c is open.

Given a set, we characterize its inside, the smallest closed set containing it, as well as its boundary.

Definition 1.3.4. Let X be a metric space and $E \subset X$.

- (1) The *interior of E* is the set:

$$E^\circ := \bigcup \{U \subset E \mid U \text{ is open}\}.$$

(2) The *closure* of E is the set:

$$\bar{E} := \bigcap \{C \supset E \mid C \text{ is closed}\}.$$

(3) The *boundary* of E is the set:

$$\partial E = \bar{E} \setminus E^\circ.$$

Proposition 1.3.1. *A countable subset of $\mathbb{R}$ has an empty interior.*

Proof. Let A be a subset of $\mathbb{R}$ which has a nonempty interior. Then A contains a non-trivial interval. Hence $\text{card}(A) \geq \aleph_0$. Thus A is uncountable. $\square$

Example 1.3.1. Let $E = \mathbb{Q} \cap (0, 1)$. We have $E^\circ = \emptyset$ and $\bar{E} = [0, 1]$.

Definition 1.3.5. Let X be a metric space and $E \subset X$.

- (1) E is *dense* in X if $\bar{E} = X$.
- (2) E is *nowhere dense* if $(\bar{E})^\circ = \emptyset$.

Intuitively, a set is nowhere dense if we can't fit an open ball inside it.

Definition 1.3.6. Let (X, d) be a metric space.

- (1) The collection of open sets of X is denoted τ_X .
- (2) A *base* for τ_X is a family of open subsets $\mathcal{B} \subseteq \tau_X$ such that:

$$(\forall U \in \tau_X)(\forall x \in U)(\exists B \in \mathcal{B}) : x \in B \subseteq U.$$

Equivalently, for all $U \in \tau_X$, we can write $U = \bigcup_{i \in I} B_i$, where $\{B_i\}_{i \in I} \subseteq \mathcal{B}$.

- (3) X is *second countable* if it has a countable base.

Definition 1.3.7. Let (X, ρ) be a metric space. X is *separable* if there exists a countable dense subset.

Proposition 1.3.2. *** *Let (X, ρ) be a metric space. X is second countable if and only if X is separable.*

Proof. $\square$

With a metric we can talk about the distance from a point to a set, as well as a way to define if a set is bounded.

Definition 1.3.8. Let (X, ρ) be a metric space, $x \in X$, and $E \subset X$.

- (1) The distance from x to E is $\rho(x, E) := \inf_{y \in E} \rho(x, y)$.
- (2) The *diameter* of E is $\text{diam}(E) = \sup_{x, y \in E} \rho(x, y)$.
- (3) A subset of a metric space is *bounded* if its diameter is finite.

With a notion of distance, we can talk about sequence convergence.

Definition 1.3.9. Let (X, ρ) be a metric space and $(x_n)_n$ a sequence in X . We say $(x_n)_n$ converges to a point x_0 if $\lim_{n \rightarrow \infty} \rho(x_n, x_0) = 0$.

Proposition 1.3.3. Let (X, ρ) be a metric space, $E \subset X$, and $x \in X$. The following are equivalent:

- (1) $x \in \bar{E}$;
- (2) For every $r > 0$, $B(r, x) \cap E \neq \emptyset$;
- (3) There is a sequence $(x_n)_n$ in E which converges to x

We can even talk about continuity of functions between metric spaces.

Definition 1.3.10. Let (X_1, ρ_1) and (X_2, ρ_2) be metric spaces and $f : X_1 \rightarrow X_2$.

- (1) We say f is *continuous* if:

$$(\forall c \in X_1)(\forall \epsilon > 0)(\exists \delta > 0) : (\forall x \in X_1)(\rho_1(x, c) < \delta \implies \rho_2(f(x), f(c)) < \epsilon).$$

- (2) We say f is *uniformly continuous* if:

$$(\forall \epsilon > 0)(\exists \delta > 0) : (\forall x, y \in X_1)(\rho_1(x, y) < \delta \implies \rho_2(f(x), f(y)) < \epsilon).$$

- (3) We say f is *Lipschitz* if:

$$(\exists C \geq 0) : (\forall x, y \in X_1)(\rho_2(f(x), f(y)) \leq C \cdot \rho_1(x, y)).$$

Note that δ may depend on our point c in the case that f is continuous. A function is uniformly continuous if δ does not depend on any two points in its domain.

Proposition 1.3.4. Let (X_1, ρ_1) and (X_2, ρ_2) be metric spaces. f is continuous if and only if the preimage of every open set in X_2 is open in X_1 .

Similarly to how every Cauchy sequence in $\mathbb{R}$ converges, we can talk about metric spaces which guarantee that same property.

Definition 1.3.11.

- (1) Let $(x_n)_n$ be a sequence in a metric space (X, ρ) . We say $(x_n)_n$ is ρ -*Cauchy* if:

$$(\forall \epsilon > 0)(\exists N \in \mathbb{N}) : (\forall n, m \in \mathbb{N})(n > m \geq N \implies \rho(x_n, x_m) < \epsilon).$$

- (2) A metric space X is called *complete* if every Cauchy sequence converges to a point in X .

We can generalize the notion of a set being closed and bounded in $\mathbb{R}^n$.

Proposition 1.3.5. Let X be a metric space. The following are equivalent:

- (1) Every open cover of X admits a finite subcover;
- (2) Every sequence in X admits a convergent subsequence;
- (3) X is complete and totally bounded.

Definition 1.3.12. We say a metric space X is *compact* if it satisfies any of the properties in Proposition 1.3.5.

A set may have more than one metric defined on it.

Definition 1.3.13. Let X be a metric space. We say ρ_1 and ρ_2 are *equivalent metrics on X* if $\text{id} : (X, \rho_1) \rightarrow (X, \rho_2)$ and $\text{id}^{-1} : (X, \rho_2) \rightarrow (X, \rho_1)$ are Lipschitz.

Chapter 2

Measures

Ideally in $\mathbb{R}$, we would like to have a function μ that assigns each $E \subset \mathbb{R}$ to a number $\mu(E) \in [0, \infty]$. Such a function μ should possess the "essence" of length:

1. The unit cube should have a measure equal to one—that is, $\mu([0, 1)) = 1$;
2. If E and F are congruent subsets of $\mathbb{R}$ (that is, F is a translation, rotation, and/or reflection of E), then $\mu(E) = \mu(F)$;
3. If $E_1, E_2, \dots$ are pairwise disjoint, then $\mu\left(\bigcup_{j=1}^{\infty} E_j\right) = \sum_{j=1}^{\infty} \mu(E_j)$.

Unfortunately, a function possessing all of these traits does not exist.

Theorem 2.0.1. *A function $\mu : \mathcal{P}(\mathbb{R}) \rightarrow [0, \infty]$ satisfying (1), (2), and (3) from above does not exist.*

Proof. Suppose such a function μ does exist. By (1) we have $\mu([0, 1)) = 1$. Define a relation on $[0, 1)$ as follows: $x \sim y$ if and only if $x - y \in \mathbb{Q}$. This is an equivalence relation. Using the Axiom of Choice, let N be a set which contains exactly one element from each equivalence class. Now for each $q \in \mathbb{Q} \cap [0, 1)$, define:

$$N_q = \{x + q \mid x \in N \cap [0, 1 - q)\} \cup \{x + (q - 1) \mid x \in N \cap [1 - q, 1)\}.$$

That is, the points in N_q are obtained by "rotating" the points in N by a rational number q . Clearly $N_q \subset [0, 1)$, hence $\bigcup_{q \in \mathbb{Q} \cap [0, 1)} N_q \subset [0, 1)$. We have the following three properties:

- (a) For any $q \in \mathbb{Q}$, $\mu(N) = \mu(N_q)$
- (b) If $r, s \in \mathbb{Q}$, $r \neq s$, then $N_r \cap N_s = \emptyset$.
- (c) For every $y \in [0, 1)$, there exists a $q \in \mathbb{Q}$ such that $y \in N_q$.

Property (a) follows from the fact that N and N_r are congruent, hence:

$$\begin{aligned} \mu(N) &= \mu(N \cap [0, 1 - q)) + \mu(N \cap [1 - q, 1)) \\ &= \mu(\{x + q \mid x \in N \cap [0, 1 - q)\}) + \mu(\{x + q - 1 \mid x \in N \cap [1 - q, 1)\}) \quad \text{"Rotation"} \\ &= \mu(N_r). \end{aligned}$$

Property (b) follows by contradiction—if $x \in N_r \cap N_s$, then $x - r$ and $x - s$ would be distinct elements of N . However, $(x - r) - (x - s) = s - r \in \mathbb{Q}$, so $x - r$ and $x - s$ are in the same equivalence class. But this contradicts the way which we've constructed N . For property (c), given $y \in [0, 1)$, let $x \in N$ satisfy $x \sim y$. Since x and y are in the same equivalence class,

we can find $q \in \mathbb{Q}$ such that:

$$y = \begin{cases} x + q, & x < y \\ x - (q - 1) & x > y \end{cases}.$$

Hence $y \in N_q$, giving $[0, 1) \subset N_q \subset \bigcup_{q \in \mathbb{Q} \cap [0, 1)} N_q$. We've obtained the equality:

$$\bigcup_{q \in \mathbb{Q} \cap [0, 1)} N_q = [0, 1),$$

and by property (a) this becomes:

$$\bigsqcup_{q \in \mathbb{Q} \cap [0, 1)} N_q = [0, 1).$$

Finally, with property (a), we have:

$$\begin{aligned} 1 &= \mu([0, 1)) \\ &= \mu\left(\bigsqcup_{q \in \mathbb{Q} \cap [0, 1)} N_q\right) \\ &= \sum_{q \in \mathbb{Q} \cap [0, 1)} \mu(N_q) \\ &= \sum_{n=1}^{\infty} \mu(N). \end{aligned}$$

This is impossible, hence there does not exist a μ satisfying (1), (2), and (3). $\square$

One might guess that this inconsistency is a result of assuming countable additivity as opposed to finite additivity. This is not the case, and in dimensions $n \geq 3$ assuming a weak form of μ leads to the following result:

Theorem 2.0.2 (Banach-Tarski). *Let U and V be arbitrary bounded open sets in $\mathbb{R}^n$ for $n \geq 3$. There exist $k \in \mathbb{N}$ and subsets $E_1, \dots, E_k, F_1, \dots, F_k$ of $\mathbb{R}^n$ such that:*

- *the E_j 's are disjoint and their union is U ;*
- *the F_j 's are disjoint and their union is V ;*
- *E_j is congruent to F_j for all $1 \leq j \leq k$.*

Ultimately, the problem comes from the fact we defined μ on *every possible* subset of $\mathbb{R}^n$. We can't hope to do this, since as we showed there exists bizarre sets which cannot be measured. Instead, we will define μ on a special class of subsets of $\mathbb{R}^n$.

§ 2.1. σ -Algebras

Definition 2.1.1. Let X be a set. An *algebra of sets* $\mathcal{A} \subset \mathcal{P}(X)$ is a collection of subsets of X which is:

(1) closed under finite union: if $E_1, E_2, \dots, E_n \in \mathcal{A}$, then $\bigcup_{i=1}^n E_i \in \mathcal{A}$;

(2) closed under complement: if $E \in \mathcal{A}$, then $E^c \in \mathcal{A}$.

We used the ordered pair $(X, \mathcal{A})$... If $\mathcal{A}$ is closed under countable union, then we say $\mathcal{A}$ is a σ -algebra.

Proposition 2.1.1. *Let $\mathcal{A}$ be an algebra of sets.*

(1) $\mathcal{A}$ is closed under finite intersection.

(2) $\mathcal{A}$ is closed under set difference.

(3) $\emptyset \in \mathcal{A}$ and $X \in \mathcal{A}$.

Proof. (1) Let $E_1, E_2, \dots, E_n \in \mathcal{A}$. Then $\bigcap_{i=1}^n E_i = \left(\bigcup_{i=1}^n E_i^c\right)^c \in \mathcal{A}$. (2) If $E_1, E_2 \in \mathcal{A}$, then $E_1 \setminus E_2 = E_1 \cap E_2^c \in \mathcal{A}$. (3) For any $E \in \mathcal{A}$, we have $\emptyset = E \cap E^c \in \mathcal{A}$ and $X = E \cup E^c \in \mathcal{A}$. $\square$

Proposition 2.1.2. *Let $\mathcal{A}$ be an algebra of sets. The following are equivalent:*

(1) $\mathcal{A}$ is a σ -algebra.

(2) $\mathcal{A}$ is closed under countable disjoint union.

Proof. If $\mathcal{A}$ is a σ -algebra, then it is closed under countable union, hence it must be closed under countable disjoint union. Conversely, suppose $\mathcal{A}$ is closed under countable disjoint union. Let $\{E_n\}_{n=1}^\infty$ be a family of sets in $\mathcal{A}$. Define:

$$\begin{aligned} F_1 &= E_1, \\ F_2 &= E_2 \setminus E_1 \\ F_3 &= E_3 \setminus (E_1 \cup E_2) \\ &\vdots \\ F_n &= E_n \setminus (E_1 \cup E_2 \cup \dots \cup E_{n-1}). \end{aligned}$$

Let $i < j$ be natural numbers and consider $F_i \cap F_j$. If $x \in F_i$, then $x \in E_i$. Since $i < j$, we have $x \in E_1 \cup \dots \cup E_{j-1}$, hence $x \notin F_j$. Thus $F_i \cap F_j = \emptyset$. Inductively, $\{F_n\}_{n=1}^\infty$ is a countable family of disjoint sets.

Clearly $\bigcup_{n=1}^\infty F_n \subset \bigcup_{n=1}^\infty E_n$. Conversely, let $x \in \bigcup_{n=1}^\infty E_n$. Find the smallest natural number j such that $x \in E_j$. Then $x \notin E_i$ for $i < j$. Hence $x \in E_j \setminus (E_1 \cup \dots \cup E_{j-1}) = F_j \subset \bigcup_{n=1}^\infty F_n$.

Since $\mathcal{A}$ is closed under countable disjoint unions, we have $\bigcup_{n=1}^\infty E_n = \bigcup_{n=1}^\infty F_n \in \mathcal{A}$. Since $\{E_n\}_{n=1}^\infty$ was an arbitrary family of sets, this means $\mathcal{A}$ is a σ -algebra. $\square$

Proposition 2.1.3. *If $\{\mathcal{E}_i\}_{i \in I}$ is a family of σ -algebras, then $\bigcap_{i \in I} \mathcal{E}_i$ is a σ -algebra.*

Proof. Let $\{E_n\}_{n=1}^\infty$ be a family of sets contained in $\bigcap_{i \in I} \mathcal{E}_i$. Then each $\{E_n\}_{n=1}^\infty$ is contained in $\mathcal{E}_i$. Since $\mathcal{E}_i$ is a σ -algebra for every $i \in I$, we have $\bigcup_{n=1}^\infty E_n \in \mathcal{E}_i$ for every $i \in I$, which means $\bigcup_{n=1}^\infty E_n \in \bigcap_{i \in I} \mathcal{E}_i$.

Similarly, let $F \in \bigcap_{i \in I} \mathcal{E}_i$. Then $F \in \mathcal{E}_i$ for every $i \in I$. Since $\mathcal{E}_i$ is a σ -algebra for every $i \in I$, we have $F^c \in \mathcal{E}_i$ for every $i \in I$, which means $F^c \in \bigcap_{i \in I} \mathcal{E}_i$. Thus $\bigcap_{i \in I} \mathcal{E}_i$ is a σ -algebra. $\square$

Definition 2.1.2. Let X be a set and $\mathcal{E} \subset \mathcal{P}(X)$. The σ -algebra generated by $\mathcal{E}$ is $\mathfrak{M}(\mathcal{E}) := \bigcap \{\mathcal{A} \mid \mathcal{E} \subset \mathcal{A}, \mathcal{A} \text{ is a } \sigma\text{-algebra}\}$.

Lemma 2.1.4. *If $\mathcal{E} \subset \mathfrak{M}(\mathcal{F})$, then $\mathfrak{M}(\mathcal{E}) \subset \mathfrak{M}(\mathcal{F})$.*

Proof. Since $\mathfrak{M}(\mathcal{F})$ is a σ -algebra containing $\mathcal{E}$, it must contain $\mathfrak{M}(\mathcal{E})$, as $\mathfrak{M}(\mathcal{E})$ is the smallest σ -algebra generated by $\mathcal{E}$. $\square$

For a metric space X , it'd be nice if we could measure all the open sets.

Definition 2.1.3. Let X be a metric space and τ_X the set containing all of its open sets. The *Borel σ -algebra on X* is the set $\mathcal{B}_X := \mathfrak{M}(\tau_X)$. Members of $\mathcal{B}_X$ are called *Borel sets*.

Exposition on appreciating whats inside the borel sets. define was F sigma and G delta sets are. talk about why they are in it and also combinations of them

Proposition 2.1.5. *$\mathcal{B}_{\mathbb{R}}$ is generated by each of the following:*

- (1) *the open intervals:* $\mathcal{E}_1 = \{(a, b) \mid a < b\}$,
- (2) *the closed intervals:* $\mathcal{E}_2 = \{[a, b] \mid a < b\}$,
- (3) *the half-open intervals:* $\mathcal{E}_3 = \{(a, b] \mid a < b\}$ or $\mathcal{E}_4 = \{[a, b) \mid a < b\}$,
- (4) *the open rays:* $\mathcal{E}_5 = \{(a, \infty) \mid a \in \mathbb{R}\}$ or $\mathcal{E}_6 = \{(-\infty, a) \mid a \in \mathbb{R}\}$,
- (5) *the closed rays:* $\mathcal{E}_7 = \{(a, \infty) \mid a \in \mathbb{R}\}$ or $\mathcal{E}_8 = \{(-\infty, a) \mid a \in \mathbb{R}\}$.

Proof. We are required to show for each $\mathcal{E}_i$ that $\mathfrak{M}(\mathcal{E}_i) = \mathcal{B}_{\mathbb{R}}$.

(1) Let $I \in \mathcal{E}_1$ be an open interval. Then $I \in \mathcal{B}_{\mathbb{R}}$ because I is an open subset of $\mathbb{R}$. From this we have $\mathcal{E}_1 \subset \mathcal{B}_{\mathbb{R}}$, but because $\mathcal{B}_{\mathbb{R}}$ is a σ -algebra containing $\mathcal{E}_1$, and $\mathfrak{M}(\mathcal{E}_1)$ is the smallest σ -algebra containing $\mathcal{E}_1$, we have $\mathfrak{M}(\mathcal{E}_1) \subset \mathcal{B}_{\mathbb{R}}$. Conversely, let $U \in \tau_{\mathbb{R}}$. Then U is at most the countable union of disjoint open intervals, hence $U \in \mathfrak{M}(\mathcal{E}_1)$. From this we have $\tau_{\mathbb{R}} \subset \mathfrak{M}(\mathcal{E}_1)$, but because $\mathfrak{M}(\mathcal{E}_1)$ is a σ -algebra containing $\tau_{\mathbb{R}}$, and $\mathfrak{M}(\tau_{\mathbb{R}})$ is the smallest σ -algebra containing $\tau_{\mathbb{R}}$, we have $\mathfrak{M}(\tau_{\mathbb{R}}) = \mathcal{B}_{\mathbb{R}} \subset \mathfrak{M}(\mathcal{E}_1)$.

(2) Let $I \in \mathcal{E}_1$ be an open interval. Then I^c is a closed interval, hence it is contained in $\mathfrak{M}(\mathcal{E}_2)$. Whence $\mathfrak{M}(\mathcal{E}_1) \subset \mathfrak{M}(\mathcal{E}_2)$. Conversely, let $C \in \mathcal{E}_2$ be a closed interval. Then C can be written as the countable intersection of open sets. This means $C \in \mathfrak{M}(\mathcal{E}_1)$. Whence $\mathfrak{M}(\mathcal{E}_2) \subset \mathfrak{M}(\mathcal{E}_1)$. Thus $\mathfrak{M}(\mathcal{E}_2) = \mathfrak{M}(\mathcal{E}_1) = \mathcal{B}_{\mathbb{R}}$ $\square$

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Given two sets X_1 and X_2 with respective σ -algebras $\mathcal{M}_1$ and $\mathcal{M}_2$, we'd like to better understand what a σ -algebra on $X_1 \times X_2$ looks like. Complications arise depending on the cardinality of our index set, so for this reason it helps to see the *categorical* definition of what products of sets look like.

Definition 2.1.4. Let $\{X_\alpha\}_{\alpha \in I}$ be a family of objects indexed by a set I . A *product* of the family is an object X equipped with an I -indexed family of morphisms $\pi_\alpha : X \rightarrow X_\alpha$ satisfying the following universal property: for every object Y and every I -indexed family of morphisms $f_\alpha : Y \rightarrow X_\alpha$, there exists a unique morphism $f : Y \rightarrow X$ such that the following diagram commutes for every $\alpha \in I$:

$$\begin{array}{ccc}
 & & X \\
 & \nearrow f & \downarrow \pi_\alpha \\
 Y & \dashrightarrow f_\alpha & X_\alpha
 \end{array}$$

In Category theory, these products are *only* defined by this universal property. Essentially, the product of a family of objects is the "most general" object which admits a morphism to each of the given objects. In the category of sets, this product is the *Cartesian product*, which we've defined earlier.

Definition 2.1.5. Let $\{X_\alpha\}_{\alpha \in I}$ be an indexed family of nonempty sets, $X = \prod_{\alpha \in I} X_\alpha$, and $\pi_\alpha : X \rightarrow X_\alpha$ the projection maps. If $\mathcal{M}_\alpha$ is a σ -algebra on X_α for each α , the *product σ -algebra* on X is:

$$\bigotimes_{\alpha \in I} \mathcal{M}_\alpha := \mathfrak{M}\left(\{\pi_\alpha^{-1}(E_\alpha) \mid E_\alpha \in \mathcal{M}_\alpha, \alpha \in I\}\right)$$

The significance of this definition will become clearer later. If X_1 and X_2 are sets with σ -algebras $\mathcal{M}_1$ and $\mathcal{M}_2$ respectively, one might wager a more intuitive definition for the σ -algebra on $X_1 \times X_2$ to be:

$$\mathfrak{M}\left(\{E_1 \times E_2 \mid E_1 \in \mathcal{M}_1, E_2 \in \mathcal{M}_2\}\right).$$

In fact, these two definitions are equivalent for the case where we take the cartesian product of countably many sets! Before presenting this as a proposition, note that if $E_i \subset X_i$ for $i = 1, 2$ and $\pi_i : X_1 \times X_2 \rightarrow X_i$ then:

$$\begin{aligned}
 \pi_1^{-1}(E_1) &= E_1 \times X_2, \\
 \pi_2^{-1}(E_2) &= X_1 \times E_2.
 \end{aligned}$$

We can see that:

$$\begin{aligned}
 \pi_1^{-1}(E_1) \cap \pi_2^{-1}(E_2) &= (E_1 \times X_2) \cap (X_1 \times E_2) \\
 &= \{(e_1, x_2) \mid e_1 \in E_1, x_2 \in X_2\} \cap \{(x_1, e_2) \mid x_1 \in X_1, e_2 \in E_2\} \\
 &= \{(x, y) \mid (x \in E_1 \wedge x \in X_1) \wedge (y \in X_2 \wedge y \in E_2)\} \\
 &= \{(x, y) \mid x \in E_1 \wedge y \in E_2\} \\
 &= E_1 \times E_2.
 \end{aligned}$$

Proposition 2.1.6. *** *this could be cleaned up* Let A be countable, $\{X_\alpha\}_{\alpha \in A}$ an indexed family of nonempty sets, $X = \prod_{\alpha \in A} X_\alpha$, $\pi_\alpha : X \rightarrow X_\alpha$ the projection maps, and $\mathcal{M}_\alpha$ a σ -algebra on X_α for each $\alpha \in A$. Then $\bigotimes_{\alpha \in A} \mathcal{M}_\alpha$ is generated by:

$$\left\{ \prod_{\alpha \in A} E_\alpha \mid E_\alpha \in \mathcal{M}_\alpha \right\}$$

Proof. Denote $\mathcal{O}$ as the σ -algebra generated by $\{\prod_{\alpha \in A} E_\alpha \mid E_\alpha \in \mathcal{M}_\alpha\}$. If $E_\alpha \in \mathcal{M}_\alpha$, then $\pi_\alpha^{-1}(E_\alpha) = \prod_{\beta \in A} E_\beta$, where $E_\beta = X_\beta$ for $\beta \neq \alpha$. This gives the inclusion:

$$\{\pi_\alpha^{-1}(E_\alpha) \mid E_\alpha \in \mathcal{M}_\alpha, \alpha \in I\} \subset \mathcal{O}.$$

But since $\bigotimes_{\alpha \in A} \mathcal{M}_\alpha$ is the smallest σ -algebra generated by $\{\pi_\alpha^{-1}(E_\alpha) \mid E_\alpha \in \mathcal{M}_\alpha, \alpha \in I\}$, we get:

$$\bigotimes_{\alpha \in A} \mathcal{M}_\alpha \subset \mathcal{O}.$$

Conversely, note that $\prod_{\alpha \in A} E_\alpha = \bigcap_{\alpha \in A} \pi_\alpha^{-1}(E_\alpha)$. Since each $\pi_\alpha^{-1}(E_\alpha)$ is in $\bigotimes_{\alpha \in A} \mathcal{M}_\alpha$, and since $\bigotimes_{\alpha \in A} \mathcal{M}_\alpha$ is closed under countable intersections, we have that $\prod_{\alpha \in A} E_\alpha \in \bigotimes_{\alpha \in A} \mathcal{M}_\alpha$. Hence we have the following inclusion:

$$\left\{ \prod_{\alpha \in A} E_\alpha \mid E_\alpha \in \mathcal{M}_\alpha \right\} \subset \bigotimes_{\alpha \in A} \mathcal{M}_\alpha.$$

But since $\mathcal{O}$ is the smallest σ -algebra generated by $\{\prod_{\alpha \in A} E_\alpha \mid E_\alpha \in \mathcal{M}_\alpha\}$, it must be the case that $\mathcal{O} \subset \bigotimes_{\alpha \in A} \mathcal{M}_\alpha$. Since we showed inclusion in both directions, $\mathcal{O} = \bigotimes_{\alpha \in A} \mathcal{M}_\alpha$; i.e., $\bigotimes_{\alpha \in A} \mathcal{M}_\alpha$ is generated by $\{\prod_{\alpha \in A} E_\alpha \mid E_\alpha \in \mathcal{M}_\alpha\}$. $\square$

In both of the above cases, we do not assume each $\mathcal{M}_\alpha$ to be generated by any set. If we assume that each $\mathcal{M}_\alpha$ to be generated by a set, we have a

Lemma 2.1.7. *** Let $\{X_\alpha\}_{\alpha \in A}$ be a family of nonempty sets, $\mathcal{E}_\alpha \subset \mathcal{P}(X_\alpha)$, and $\mathcal{M}_\alpha := \mathfrak{M}(\mathcal{E}_\alpha)$ for each $\alpha \in A$. Then $\bigotimes_{\alpha \in A} \mathcal{M}_\alpha$ is generated by:

$$\mathcal{F}_1 = \{\pi_\alpha^{-1}(E_\alpha) \mid E_\alpha \in \mathcal{E}_\alpha, \alpha \in A\}.$$

If A is countable and $X_\alpha \in \mathcal{E}_\alpha$ for each α , then $\bigotimes_{\alpha \in A} \mathcal{M}_\alpha$ is generated by:

$$\mathcal{F}_2 = \left\{ \prod_{\alpha \in A} E_\alpha \mid E_\alpha \in \mathcal{E}_\alpha \right\}.$$

Proof. Automatically we have $\mathcal{F}_1 \subset \bigotimes_{\alpha \in A} \mathcal{M}_\alpha$, hence $\mathfrak{M}(\mathcal{F}_1) \subset \bigotimes_{\alpha \in A} \mathcal{M}_\alpha$. Conversely, consider the set $S := \{E \subset X_\alpha \mid \pi_\alpha^{-1}(E) \in \mathfrak{M}(\mathcal{F}_1)\}$. This set is a σ -algebra. Indeed, if $\{E_n\}_{n \in \mathbb{N}}$ is a family of elements of S , then $\{\pi_\alpha^{-1}(E_n)\}_{n \in \mathbb{N}}$ is a family of elements of $\mathfrak{M}(\mathcal{F}_1)$. Since $\mathfrak{M}(\mathcal{F}_1)$ is a σ -algebra, we get:

$$\bigcup_{n \in \mathbb{N}} \pi_\alpha^{-1}(E_n) = \pi_\alpha^{-1} \left(\bigcup_{n \in \mathbb{N}} E_n \right).$$

Hence $\bigcup_{n \in \mathbb{N}} E_n \in S$. A similar argument can be used to show S is closed under complements. With this in mind, note that we also have $\mathcal{E}_\alpha \subset \mathfrak{M}(\mathcal{F}_1)$. Indeed, for any $E \in \mathcal{E}_\alpha$, $\square$

Corollary 2.1.8. *** Let $X_1, \dots, X_n$ be metric spaces and let $X = \prod_{j=1}^n X_j$, equipped with the product metric. Then $\bigotimes_{j=1}^n \mathcal{B}_{X_j} \subset \mathcal{B}_X$. If the X_j 's are separable, then $\bigotimes_{j=1}^n \mathcal{B}_{X_j} = \mathcal{B}_X$.

Proof. I need to make sure to do this. $\square$

Corollary 2.1.9. *** $\mathcal{B}_{\mathbb{R}^n} = \bigotimes_{j=1}^n \mathcal{B}_{\mathbb{R}}$.

§ 2.2. Measures

Definition 2.2.1. Let X be a nonempty set and $\mathcal{M}$ a σ -algebra on X . A *measure* is a function:

$$\mu : \mathcal{M} \rightarrow [0, \infty]$$

satisfying:

(1) $\mu(\emptyset) = 0$;

(2) if $\{E_j\}_{j=1}^{\infty} \subset \mathcal{M}$ are pairwise disjoint, then $\mu\left(\bigcup_{j=1}^{\infty} E_j\right) = \sum_{j=1}^{\infty} \mu(E_j)$.

We call the ordered triple $(X, \mathcal{M}, \mu)$ a *measure space*. The sets $E \subset \mathcal{M}$ are called *measurable sets*. If $\mu(X) < \infty$, then μ is called *finite*. If $X = \bigcup_{j=1}^{\infty} E_j$, $E_j \in \mathcal{M}$ for all j , and $\mu(E_j) < \infty$, then μ is *σ -finite*.

Example 2.2.1.

- (1) Let X be a nonempty set. Define $\mu : \mathcal{P}(X) \rightarrow [0, \infty]$ by $\mu(E) = \sum_{x \in E} 1$ (note that this series exists as our codomain is the extended real numbers). Then $(X, \mathcal{P}(X), \mu)$ is a measure space, and we call μ the *counting measure*.
- (2) Let X be nonempty and fix a point $x_0 \in X$. Define $\mu : \mathcal{P}(X) \rightarrow [0, \infty]$ by:

$$\mu(E) = \begin{cases} 0, & x_0 \notin E \\ 1, & x_0 \in E. \end{cases}$$

Then $(X, \mathcal{M}, \mu)$ is a measure space, and we call μ the *dirac measure*.

- (3) Examples (1) and (2) can be generalized. Let X be a nonempty set and $f : X \rightarrow [0, \infty]$. Define $\mu : \mathcal{P}(X) \rightarrow [0, \infty]$ by $\mu(E) = \sum_{x \in E} f(x)$. Then $(X, \mathcal{P}(X), \mu)$ is a measure space. If $f : X \rightarrow [0, \infty]$ is the identity function, then μ is the counting measure. If $f : X \rightarrow [0, \infty]$ is the indicator function $\mathbb{1}_{\{x_0\}}$, then μ is the dirac measure.
- (4) Let X be an uncountable set and let $\mathcal{M}$ be the σ -algebra of countable or co-countable sets. Define $\mu : \mathcal{M} \rightarrow [0, \infty]$ by:

$$\mu(E) = \begin{cases} 0, & E \text{ is countable} \\ 1, & E \text{ is co-countable.} \end{cases}$$

Then $(X, \mathcal{M}, \mu)$ is a measure space.

Exercise 2.2.1. Let $(X, \mathcal{M}, \mu)$ be a measure space and $E, F \in \mathcal{M}$. Then $\mu(E) + \mu(F) = \mu(E \cup F) + \mu(E \cap F)$.

Proof. (1) We can write $E \cup F = (E \setminus F) \sqcup (E \cap F) \sqcup (F \setminus E)$. This means $\mu(E \cup F) = \mu(E \setminus F) + \mu(E \cap F) + \mu(F \setminus E)$. This is equivalent to:

$$\begin{aligned} \mu(E \cup F) &= \left(\mu(E \setminus F) + \mu(E \cap F) \right) + \left(\mu(F \setminus E) + \mu(E \cap F) \right) - \mu(E \cap F) \\ &= \mu\left((E \setminus F) \cup (E \cap F)\right) + \mu\left((F \setminus E) \cup (E \cap F)\right) - \mu(E \cap F) \\ &= \mu(E) + \mu(F) - \mu(E \cap F) \end{aligned}$$

Hence $\mu(E) + \mu(F) = \mu(E \cup F) + \mu(E \cap F)$. □

Proposition 2.2.1. *Let $(X, \mathcal{M}, \mu)$ be a measure space.*

- (1) *If $E, F \in \mathcal{M}$ and $E \subset F$, then $\mu(E) \leq \mu(F)$ (Monotonicity).*
- (2) *Let $\{E_i\}_{i=1}^\infty \subset \mathcal{M}$. Then $\mu\left(\bigcup_{i=1}^\infty E_i\right) \leq \sum_{i=1}^\infty \mu(E_i)$ (Subadditivity).*
- (3) *Let $\{E_i\}_{i=1}^\infty \subset \mathcal{M}$ with $E_1 \subset E_2 \subset E_3 \subset \dots$. Then $\mu\left(\bigcup_{i=1}^\infty E_i\right) = \lim_{i \rightarrow \infty} \mu(E_i)$ (Continuity from below).*
- (4) *Let $\{E_i\}_{i=1}^\infty \subset \mathcal{M}$ with $E_1 \supset E_2 \supset E_3 \supset \dots$ provided $\mu(E_1) < \infty$. Then $\mu\left(\bigcap_{i=1}^\infty E_i\right) = \lim_{i \rightarrow \infty} \mu(E_i)$.*

Proof. (1) We can write $F = E \sqcup (F \setminus E)$, giving $\mu(F) = \mu(E \sqcup (F \setminus E)) = \mu(E) + \mu(F \setminus E) \geq \mu(E)$.

(2) Given $\{E_i\}_{i=1}^\infty \subset \mathcal{M}$, define $F_i = E_i \setminus \bigcup_{k=1}^{i-1} E_k$ for each i . Then $\{F_i\}_{i=1}^\infty$ is a sequence of pairwise disjoint subsets of X . In particular, we have $F_i \subset E_i$ for each i , and $\bigcup_{i=1}^\infty E_i = \bigcup_{i=1}^\infty F_i$. Together:

$$\begin{aligned} \mu\left(\bigcup_{i=1}^\infty E_i\right) &= \mu\left(\bigcup_{i=1}^\infty F_i\right) \\ &= \sum_{i=1}^\infty \mu(F_i) \\ &\leq \sum_{i=1}^\infty \mu(E_i). \end{aligned}$$

(3) Define $E_0 = \emptyset$. Then $\{E_i \setminus E_{i-1}\}_{i=1}^\infty$ will be a sequence of pairwise disjoint subsets of X with the property that $\bigcup_{i=1}^\infty E_i = \bigcup_{i=1}^\infty E_i \setminus E_{i-1}$. With this:

$$\begin{aligned} \mu\left(\bigcup_{i=1}^\infty E_i\right) &= \mu\left(\bigcup_{i=1}^\infty E_i \setminus E_{i-1}\right) \\ &= \sum_{i=1}^\infty \mu(E_i \setminus E_{i-1}). \end{aligned}$$

Note that $E_i = E_{i-1} \sqcup (E_i \setminus E_{i-1})$. This means $\mu(E_i) = \mu(E_{i-1}) + \mu(E_i \setminus E_{i-1})$, which is equivalent to $\mu(E_i \setminus E_{i-1}) = \mu(E_i) - \mu(E_{i-1})$. Hence:

$$\begin{aligned} \mu\left(\bigcup_{i=1}^\infty E_i\right) &= \sum_{i=1}^\infty \mu(E_i \setminus E_{i-1}) \\ &= \lim_{i \rightarrow \infty} \sum_{n=1}^i \mu(E_i \setminus E_{i-1}) \\ &= \lim_{i \rightarrow \infty} \sum_{n=1}^i \mu(E_i) - \mu(E_{i-1}) \\ &= \lim_{i \rightarrow \infty} \mu(E_i). \end{aligned}$$

(4) Define $F_i = E_1 \setminus E_i$. Then $\{F_i\}_{i=1}^\infty$ will be a sequence of increasing unions of subsets of X . In particular, we have that $\bigcup_{i=1}^\infty F_i = E_1 \setminus \bigcap_{i=1}^\infty E_i$. This means:

$$\begin{aligned} E_1 &= \left(E_1 \setminus \bigcap_{i=1}^\infty E_i \right) \sqcup \left(\bigcap_{i=1}^\infty E_i \right) \\ &= \left(\bigcup_{i=1}^\infty F_i \right) \sqcup \left(\bigcap_{i=1}^\infty E_i \right). \end{aligned}$$

Hence by (3):

$$\begin{aligned} \mu \left(\bigcap_{i=1}^\infty E_i \right) &= \mu(E_1) - \mu \left(\bigcup_{i=1}^\infty F_i \right) \\ &= \mu(E_1) - \lim_{i \rightarrow \infty} \mu(F_i). \end{aligned}$$

But $F_i = E_1 \setminus E_i$, so $E_1 = F_i \sqcup E_i$. This gives $\mu(E_1) = \mu(F_i) + \mu(E_i)$, hence:

$$\begin{aligned} \mu \left(\bigcap_{i=1}^\infty E_i \right) &= \mu(E_1) - \lim_{i \rightarrow \infty} \mu(F_i) \\ &= \mu(E_1) - \lim_{i \rightarrow \infty} (\mu(E_1) - \mu(E_i)) \\ &= \lim_{i \rightarrow \infty} \mu(E_i). \end{aligned} \quad \square$$

Definition 2.2.2. Let $(X, \mathcal{M}, \mu)$ be a measure space.

- (1) If $E \in \mathcal{M}$ is a measurable set such that $\mu(E) = 0$, then E is called a *null-set*.
- (2) If a property holds on all of X except on a null-set, then it holds *almost everywhere* (abbreviated *a.e.*).
- (3) If $\mathcal{M}$ has the property that every subset of a null-set is measurable, then $\mathcal{M}$ is said to be *complete*.

Theorem 2.2.2. Let $(X, \mathcal{M}, \mu)$ be a measure space. Let $\mathcal{N} := \{E \in \mathcal{M} \mid \mu(E) = 0\}$. Then the set $\overline{\mathcal{M}} := \{E \cup F \mid E \in \mathcal{M}, F \subset N \text{ for some } N \in \mathcal{N}\}$ is a σ -algebra. Moreover, there exists a unique function $\bar{\mu} : \overline{\mathcal{M}} \rightarrow [0, \infty]$ which extends μ .

Proof. Consider the family $\{E_i \cup F_i\}_{i=1}^\infty \subset \overline{\mathcal{M}}$, where for each i , there exists $N_i \in \mathcal{N}$ such that $F_i \subset N_i$. Then:

$$\bigcup_{i=1}^\infty E_i \cup F_i = \left(\bigcup_{i=1}^\infty E_i \right) \cup \left(\bigcup_{i=1}^\infty F_i \right);$$

Since $\mathcal{M}$ is a σ -algebra, we have $\bigcup_{i=1}^\infty E_i \in \mathcal{M}$. Moreover, $\bigcup_{i=1}^\infty F_i \subset \bigcup_{i=1}^\infty N_i$. By subadditivity, we have $\mu \left(\bigcup_{i=1}^\infty N_i \right) \leq \sum_{i=1}^\infty \mu(N_i) = 0$. Hence $\bigcup_{i=1}^\infty N_i \in \mathcal{N}$. Together, this shows that $\bigcup_{i=1}^\infty E_i \cup F_i \in \overline{\mathcal{M}}$.

Now let $E \cup F \in \overline{\mathcal{M}}$. Then:

$$\begin{aligned} (E \cup F)^c &= \left((E \cup N) \cap (N^c \cup F) \right)^c \\ &= (E \cup N)^c \cup (N^c \cup F)^c \\ &= (E^c \cap N^c) \cup (N \cap F^c) \\ &= (E^c \cap N^c) \cup (N \setminus F). \end{aligned}$$

Since $E^c, N^c \in \mathcal{M}$, and since $\mathcal{M}$ is closed under countable intersections, we have $E^c \cap N^c \in \mathcal{M}$. Since $N \in \mathcal{N}$, surely $N \setminus F \in \mathcal{N}$, whence $(E \cup F)^c \in \overline{\mathcal{M}}$. Thus $\overline{\mathcal{M}}$ is a σ -algebra.

Define $\bar{\mu} : \overline{\mathcal{M}} \rightarrow [0, \infty]$ by $\bar{\mu}(E \cup F) = \mu(E)$. We must first show that this map is well-defined. Suppose $E_1 \cup F_1 = E_2 \cup F_2$, where $F_i \subset N_i \in \mathcal{N}$ for $i = 1, 2$. Then $E_1 \cup F_1 \subset E_2 \cup N_2$, hence $E_1 \subset E_2 \cup N_2$. This means $\mu(E_1) \leq \mu(E_2 \cup N_2) \leq \mu(E_2) + \mu(N_2) = \mu(E_2)$. But an identical argument gives $E_2 \subset E_1 \cup N_1$, hence $\mu(E_2) \leq \mu(E_1 \cup N_1) \leq \mu(E_1) + \mu(N_1) = \mu(E_1)$. Together, $\mu(E_1) = \mu(E_2)$, allowing us to show that:

$$\begin{aligned} \bar{\mu}(E_1 \cup F_1) &= \mu(E_1) \\ &= \mu(E_2) \\ &= \bar{\mu}(E_2 \cup F_2). \end{aligned}$$

Thus $\bar{\mu}$ is well-defined. Note that $\emptyset \in \mathcal{M}$ and $\emptyset \in \mathcal{N}$, hence $\bar{\mu}(\emptyset) = \mu(\emptyset) = 0$. Now if $\{E_i \cup F_i\}_{i=1}^\infty \subset \overline{\mathcal{M}}$ are pairwise disjoint, then it must be the case that $\{E_i\}_{i=1}^\infty$ are pairwise disjoint. This gives:

$$\bar{\mu} \left(\bigcup_{i=1}^\infty E_i \cup F_i \right) = \bar{\mu} \left(\bigcup_{i=1}^\infty E_i \cup \bigcup_{i=1}^\infty F_i \right).$$

Note that $\bigcup_{i=1}^\infty E_i \in \mathcal{M}$, and $\bigcup_{i=1}^\infty F_i \subset \bigcup_{i=1}^\infty N_i \in \mathcal{N}$, hence:

$$\begin{aligned} \bar{\mu} \left(\bigcup_{i=1}^\infty E_i \cup F_i \right) &= \bar{\mu} \left(\bigcup_{i=1}^\infty E_i \cup \bigcup_{i=1}^\infty F_i \right) \\ &= \mu \left(\bigcup_{i=1}^\infty E_i \right) \\ &\leq \sum_{i=1}^\infty \mu(E_i) \\ &= \sum_{i=1}^\infty \bar{\mu}(E_i \cup F_i). \end{aligned}$$

Thus $\bar{\mu}$ is a measure. Moreover, it is an extension of μ since for any $E \in \mathcal{M}$, $\bar{\mu}(E) = \bar{\mu}(E \cup \emptyset) = \mu(E)$. It remains to show that this extension is unique. Suppose γ is also an extension of μ onto $\overline{\mathcal{M}}$. (WTS $\gamma(\overline{\mathcal{M}}) = \bar{\mu}(\overline{\mathcal{M}})$). Let $E \cup F \in \overline{\mathcal{M}}$, where $F \subset N \in \mathcal{N}$. By subadditivity:

$$\begin{aligned} \gamma(E \cup F) &\leq \gamma(E) + \gamma(F) \\ &= \gamma(E) \\ &= \mu(E) \quad \text{Since } \gamma|_{\mathcal{M}} = \mu \\ &= \bar{\mu}(E \cup F). \end{aligned}$$

But by monotonicity:

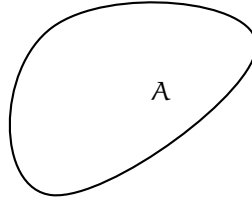
$$\begin{aligned}\bar{\mu}(E \cup F) &= \mu(E) \\ &= \gamma(E) \quad \text{Since } \gamma|_{\mathcal{M}} = \mu \\ &\leq \gamma(E \cup F).\end{aligned}$$

Hence $\gamma(E \cup F) = \bar{\mu}(E \cup F)$. Since γ and $\bar{\mu}$ are equal on all values of $\overline{\mathcal{M}}$, they are equal. Thus $\bar{\mu}$ is unique. $\square$

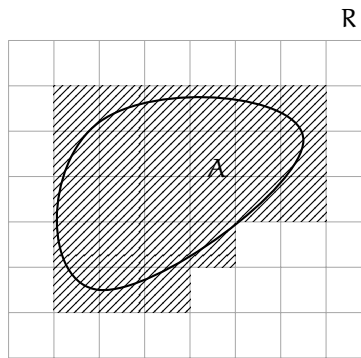
Definition 2.2.3. Let $(X, \mathcal{M}, \mu)$ be a measure space. The measure $\bar{\mu}$ is called the *completion* of μ , and $\overline{\mathcal{M}}$ is called the *completion of $\mathcal{M}$ with respect to μ* .

§ 2.3. Outer Measures

Consider the set $A \subset \mathbb{R}^2$ below:



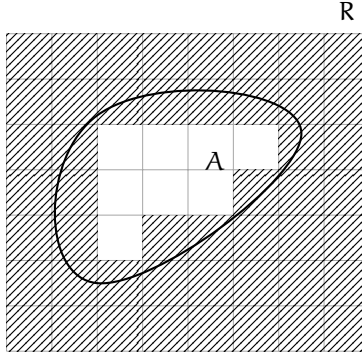
What would the "area" for this set be? Rectangles are very easy to compute, so let's cover A with an evenly partitioned rectangle R :



We can define the "outer area" of our set A as:

$$\text{outerarea}(A) = \left\{ \sum R_j \mid R_j \subset R \text{ is a rectangle covering } A \right\}.$$

Now consider the outer area of the set $R \setminus A$:



We can define the "inner area" of A as:

$$\text{innerarea}(A) = \text{area}(R) - \text{outerarea}(R \setminus A)$$

Taking finer and finer partitions, if the inner area of A is equal to the outer area of A , we can define them as the *area* of A . We can abstract this notion of outer area as follows.

Definition 2.3.1. Let X be a nonempty set. An *outer measure* on X is a function:

$$\mu^* : \mathcal{P}(X) \rightarrow [0, \infty]$$

satisfying:

- (1) $\mu^*(\emptyset) = 0$;
- (2) If $A \subset B$, then $\mu^*(A) \leq \mu^*(B)$ (Monotonicity);
- (3) $\mu^*\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} \mu^*(A_i)$ (Subadditivity).

(1) is clear otherwise we reach a contradiction. (2) follows from the fact that if we have rectangles covering B , then surely the area of those rectangles must be greater than or equal to the area of the rectangles covering A . (3) comes from the fact that if we have two overlapping sets, there may be a rectangle which covers both sets simultaneously, hence adding up the rectangles of their union is "more precise" than adding up the rectangles covering each set individually. We will prove rigorously that our motivating example does in fact give us an outer measure.

Proposition 2.3.1. Let X be any nonempty set and $\mathcal{E} \subset \mathcal{P}(X)$, with $\emptyset, X \in \mathcal{E}$. Let $\rho : \mathcal{E} \rightarrow [0, \infty]$ be a function such that $\rho(\emptyset) = 0$. The function $\mu^* : \mathcal{P}(X) \rightarrow [0, \infty]$ defined by:

$$\mu^*(A) = \inf \left\{ \sum_{i=1}^{\infty} \rho(E_i) \mid \bigcup_{i=1}^{\infty} E_i \supset A, E_i \in \mathcal{E} \right\}$$

is an outer measure on X .

Proof. Since $\emptyset \subset \emptyset$ and $\emptyset \in \mathcal{E}$, we get $\mu^*(\emptyset) = 0$. If $A \subset B$ and $\bigcup_{i=1}^{\infty} E_i \supset B$, then surely $\bigcup_{i=1}^{\infty} E_i \supset A$. This gives the containment:

$$\left\{ \sum_{i=1}^{\infty} \rho(E_i) \mid \bigcup_{i=1}^{\infty} E_i \supset B, E_i \in \mathcal{E} \right\} \supset \left\{ \sum_{i=1}^{\infty} \rho(E_i) \mid \bigcup_{i=1}^{\infty} E_i \supset A, E_i \in \mathcal{E} \right\}.$$

Any lowerbound for the left set will be a lowerbound for the right set, hence $\mu^*(A) \leq \mu^*(B)$. It remains to show that μ^* is subadditive. Let $\{A_j\}_{j=1}^\infty \subset \mathcal{P}(X)$. Let $\epsilon > 0$. For each $j = 1, 2, 3, \dots$ find a family $\{E_{i,j}\}_{i=1}^\infty \subset \mathcal{E}$ such that:

$$\sum_{i=1}^\infty \rho(E_{i,j}) \leq \frac{\epsilon}{2^j} + \mu^*(A_j). \quad \text{Property of infimums.}$$

Note that $A_j \subset \bigcup_{i=1}^\infty E_{i,j}$, hence $\{E_{i,j}\}_{i,j \in \mathbb{N}}$ covers $\bigcup_{j=1}^\infty A_j$. This means:

$$\begin{aligned} \mu^*\left(\bigcup_{j=1}^\infty A_j\right) &\leq \sum_{j=1}^\infty \sum_{i=1}^\infty \rho(E_{i,j}) \quad \text{Surely any cover will be bigger than the g.l.b.} \\ &\leq \sum_{j=1}^\infty \left(\mu^*(A_j) + \frac{\epsilon}{2^j}\right) \quad \text{Also, rearrangements of sums follows from being non-negative} \\ &= \sum_{j=1}^\infty \mu^*(A_j) + \epsilon. \end{aligned}$$

Taking $\epsilon \rightarrow \infty$ gives $\mu^*\left(\bigcup_{j=1}^\infty A_j\right) \leq \sum_{j=1}^\infty \mu^*(A_j)$. □

This next definition will lead us to constructing measures from outer measures.

Definition 2.3.2. Let μ^* be an outer measure on X . A set $A \subset X$ is called μ^* -measurable if, for all $E \subset X$:

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c).$$

This definition is very unintuitive. From our example in $\mathbb{R}^2$, given any rectangle R covering $A \subset \mathbb{R}^2$, we defined the inner area of A as:

$$\text{innerarea}(A) = \text{area}(R) - \text{outerarea}(R \setminus A),$$

where R is a rectangle covering our set A . If the inner area and the outer area are equal, then the above equation becomes:

$$\text{outerarea}(A) = \text{area}(R) - \text{outerarea}(R \setminus A).$$

Rearranging yields:

$$\text{area}(R) = \text{outerarea}(A) + \text{outerarea}(R \setminus A).$$

So under the assumption that the inner and outer area of A is equal, we gain the ability to measure any rectangle R using *only* that parts which intersect A and the parts which intersect A^c . In this case A is a "nice set" that we can use to measure other things.

Lemma 2.3.2 (Caratheodory's Lemma). **** Let μ^* be an outer measure on X . Let $\mathcal{M} = \{A \subset X \mid A \text{ is } \mu^* \text{ measurable}\}$. Then $\mathcal{M}$ is a σ -algebra, and $\mu^*|_{\mathcal{M}}$ is a complete measure (i.e., $(X, \mathcal{M}, \mu^*|_{\mathcal{M}})$ is a complete measure space).*

Proof. 10/2/25

□

Why do we care about premeasures again? Caratheodory's theorem can be used to extend measures from algebras to σ -algebra.

Definition 2.3.3. Let X be a nonempty set. Let $\mathcal{A} \subset \mathcal{P}(X)$ be an algebra of sets. A function $\mu_0 : \mathcal{A} \rightarrow [0, \infty]$ is a *premeasure* if:

- (1) $\mu_0(\emptyset) = 0$;
- (2) If $\{E_i\}_{i=1}^\infty \subset \mathcal{A}$ is a family of pairwise disjoint sets, then $\mu_0\left(\bigcup_{i=1}^\infty E_i\right) = \sum_{i=1}^\infty \mu_0(E_i)$.

Theorem 2.3.3 (Caratheodory's Extension Theorem). *** Let μ_0 be a premeasure on an algebra $\mathcal{A} \subset \mathcal{P}(X)$. Then:

- (1) There exists an extension $\mu : \mathfrak{M}(\mathcal{A}) \rightarrow [0, \infty]$ of μ_0 ;
- (2) This extension is unique if μ_0 is σ -finite.

Proof. Our first step will be to build an outer measure on $\mathcal{P}(X)$. We will use the sets from $\mathcal{A}$ as our "measuring sets". For $E \subset X$, define:

$$\mu^*(E) = \inf \left\{ \sum_{j=1}^\infty \mu_0(A_j) \mid E \subset \bigcup_{j=1}^\infty A_j, \{A_j\}_{j=1}^\infty \subset \mathcal{A} \right\}$$

Note that we can trivially cover E with $X \in \mathcal{A}$, hence it is nonempty. By Proposition 2.3.1, the function μ^* is an outer measure. By Caratheodory's Lemma, we have a measure $\mu^*|_{\mathcal{M}^*}$ defined on:

$$\mathcal{M}^* = \{A \subset X \mid A \text{ is } \mu^*\text{-measurable}\}$$

Now that we've constructed $(X, \mathcal{M}^*, \mu^*|_{\mathcal{M}^*})$, our next step is to check that $\mathcal{A} \subset \mathcal{M}^*$ —that is, for any $A \in \mathcal{A}$ and any $E \subset X$, we can write:

$$\mu^*(E) = \mu(E \cap A) + \mu(E \cap A^c).$$

Let $\epsilon > 0$. Find $\{A_j\}_{j=1}^\infty \subset \mathcal{A}$ such that:

$$\begin{aligned} \mu^*(E) + \epsilon &\geq \sum_{j=1}^\infty \mu_0(A_j) && \text{Infimum property} \\ &= \sum_{j=1}^\infty \mu_0(A_j \cap (A \cup A^c)) \\ &= \sum_{j=1}^\infty (\mu_0(A_j \cap A) + \mu_0(A_j \cap A^c)) \\ &= \sum_{j=1}^\infty \mu_0(A_j \cap A) + \sum_{j=1}^\infty \mu_0(A_j \cap A^c). \end{aligned}$$

Note that $A_j \cap A \in \mathcal{A}$ covers $E \cap A$ and $A_j \cap A^c \in \mathcal{A}$ covers $E \cap A^c$. Hence:

$$\begin{aligned} \mu^*(E) + \epsilon &\geq \sum_{j=1}^{\infty} \mu_0(A_j \cap A) + \sum_{j=1}^{\infty} \mu_0(A_j \cap A^c) && \text{Any covering will be larger than g.l.b.} \\ &\geq \mu^*(E \cap A) + \mu^*(E \cap A^c) \\ &\geq \mu^*(E). && \text{Monotonicity} \end{aligned}$$

Sending $\epsilon \rightarrow 0$ gives equality, hence A is μ^* -measurable. Having established the inclusion $\mathcal{A} \subset \mathcal{M}^*$, we obtain $\mathfrak{M}(\mathcal{A}) \subset \mathcal{M}^*$.

In summary, we constructed $(X, \mathcal{M}^*, \mu^*|_{\mathcal{M}^*})$ with respect to $\mathcal{A}$, and then showed that $\mathfrak{M}(\mathcal{A}) \subset \mathcal{M}^*$. Claim: The restriction of μ^* to $\mathfrak{M}(\mathcal{A})$ is an extension of μ_0 . For clarity, define $\mu = \mu^*|_{\mathfrak{M}(\mathcal{A})}$. 10/6/25 $\square$

§ 2.4. Measures on $\mathbb{R}$

Using the ideas from the previous section, we are now equipped to talk about measures on the real number line. As we showed, such a measure cannot be defined on the power set of $\mathbb{R}$, hence our goal is to create a measurable function defined on the Borel sets of $\mathbb{R}$. There are many options for generators of $\mathcal{B}_{\mathbb{R}}$. We will strictly use the following.

Definition 2.4.1. An *h-interval* is an interval of $\mathbb{R}$ of the form $(a, b]$ or (a, ∞) .

Note that the intersection of two h-intervals is an h-interval, and the complement of an h-interval is the union of two h-intervals. For the remainder of this section, let $\mathcal{A}$ denote the algebra of h-intervals. Then $\mathfrak{M}(\mathcal{A}) = \mathcal{B}_{\mathbb{R}}$. Using Theorem 2.3.3, if we can construct a premeasure on $\mathcal{A}$, then we can extend such a function to a measure on $\mathcal{B}_{\mathbb{R}}$.

Assume μ is a finite measure on $\mathcal{B}_{\mathbb{R}}$. Define $F : \mathbb{R} \rightarrow \mathbb{R}$ by $x \mapsto \mu((-\infty, x])$. We say F is a *distribution function*, and it has the following properties:

- F is increasing (by the monotonicity of measures);
- F is *right-continuous*.

F increasing follows from the monotonicity of measures. Now let $(x_n)_n \subset \mathbb{R}$ such that $x_n \searrow x$. Then:

$$\begin{aligned} F(x) &= \mu((-\infty, x]) \\ &= \mu\left(\bigcap_{n=1}^{\infty} (-\infty, x_n]\right) \\ &= \lim_{n \rightarrow \infty} \mu((-\infty, x_n]) && \text{Continuity from above} \\ &= F(x_n). \end{aligned}$$

Hence F is right-continuous. With these two concepts, we can see:

$$\begin{aligned} \mu((a, b]) &= \mu((-\infty, b]) - \mu((-\infty, a]) \\ &= F(b) - F(a). \end{aligned}$$

Such an outcome suggests this may be a well-defined premeasure on $\mathbb{R}$.

Proposition 2.4.1. *** Let $F : \mathbb{R} \rightarrow \mathbb{R}$ be an increasing and right-continuous function. Define $\mu_0 : \mathcal{A} \rightarrow [0, \infty]$ by $\mu_0((a, b]) = F(b) - F(a)$. Then μ_0 is a premeasure on $\mathcal{A}$ and is well-defined.

Proof. 10/7/25 □

Theorem 2.4.2. *** Let $F : \mathbb{R} \rightarrow \mathbb{R}$ be an increasing and right-continuous function. There exists a unique measure μ_F on $\mathcal{B}_{\mathbb{R}}$ such that $\mu((a, b]) = F(b) - F(a)$ for all a, b . If G is another such function, we have $\mu_F = \mu_G$ if and only if $F - G$ is constant. Conversely, if μ is a measure on $\mathcal{B}_{\mathbb{R}}$ that is finite on all bounded Borel sets adding we define:

$$F(x) = \begin{cases} \mu((0, x]), & x > 0 \\ 0, & x = 0 \\ -\mu((x, 0]), & x < 0, \end{cases}$$

then F is increasing, right continuous, and $\mu = \mu_F$.

Proof. 10/8/25 □

Note that we have the following chain of inclusions:

$$\mathcal{A} \subset \mathcal{B}_{\mathbb{R}} \subsetneq \mathcal{M}_{\mu} \subset \mathcal{P}(\mathbb{R})$$

This theorem gave us a measure μ^* on $\mathcal{M}_{\mu}$, but we only really ever cared about measures on $\mathcal{B}_{\mathbb{R}}$ since they are generated by nice sets. We got a good bang for our buck. **Are we talking about sets on $\mathcal{B}_{\mathbb{R}}$ or sets on $\mathcal{M}_{\mu}$?** Okay so basically the book is saying that $\mu^*|_{\mathcal{M}(\mathcal{A})} := \mu_F \dots$ BUT... we have the complete measure $\mu^* := \overline{\mu_F}$ on $\mathcal{M}^* := \mathcal{M}_{\mu} \dots$ Basically we are talking about properties that hold on $\mathcal{M}^* := \mathcal{M}_{\mu}$ (Carathéodory's extension theorem gave us this $\mathcal{M}^*, \dots$) Ignore what I just said and look at this table:

$$\begin{aligned} \mathcal{A} &\leftrightarrow \mu_0 \\ \mathcal{B}_{\mathbb{R}} &\leftrightarrow \mu^*|_{\mathcal{M}(\mathcal{A})} \\ \mathcal{M}_{\mu} &\leftrightarrow \mu^*|_{\mathcal{M}_{\mu}} \\ \mathcal{P}(\mathbb{R}) &\leftrightarrow \mu^* \end{aligned}$$

We're gonna be proving properties on $\mathcal{M}_{\mu}$

Definition 2.4.2. A measure satisfying the property from Proposition ??

For the remainder of this section we will fix a complete Lebesgue-Stieltjes measure μ on $\mathbb{R}$ associated to the increasing, right continuous function F . We denote by $\mathcal{M}_{\mu}$ the domain of μ . **Also, the notation in Folland is horrible**

Theorem 2.4.3. Let $E \subset \mathcal{M}_{\mu}$.

- (1) $\mu(E) = \inf\{\mu(U) \mid U \supset E, U \text{ open}\}.$
- (2) $\mu(E) = \sup\{\mu(K) \mid K \subset E, K \text{ compact}\}.$

Proof. □